

NIKHIL RAUSHAN

+91 8789120181 Bengaluru, Karnataka
nikhilraushan470@gmail.com [linkedin.com/in/nikhil-raushan-b29763252](https://www.linkedin.com/in/nikhil-raushan-b29763252)

SUMMARY

B.E. student in Electronics and Communication Engineering with hands-on experience in antenna design, digital circuit, web development, and embedded systems. Seeking an opportunity to leverage my knowledge and drive for excellence in a dynamic work environment.

EDUCATION

Bachelor of Engineering in Electronics and Communication BMS College of Engineering, Bengaluru	2022–2026
Intermediate, Trinity Global School Percentage: 82.8%	2019–2021
High School, Nezamia Public School Percentage: 77.6%	2018–2019

PROJECTS

SIW Antenna Design and Fabrication — Ansys HFSS, Microwave Design, PCB Fabrication

- Designed and fabricated a Substrate Integrated Waveguide antenna operating in X-band for high-frequency applications.
- Created parametric antenna models using Ansys HFSS to achieve optimal gain and minimal return loss.
- Improved radiation efficiency and impedance matching through iterative simulation and tuning.
- Successfully fabricated and tested the design, achieving consistent performance within lab conditions.
- Gained exposure to end-to-end RF design process—simulation, layout, and hardware testing.

Stop Clock Implementation using Verilog — Verilog, Vivado, Digital Design

- Built a stopwatch system capable of counting time intervals with start, stop, and reset functionality.
- Wrote synthesizable Verilog code for time counting, multiplexed display control, and FSM-based input logic.
- Deployed design on FPGA using Vivado; achieved real-time performance with zero timing violations.
- Modularized design blocks and optimized for area and power constraints, improving performance by 20%.
- Improved understanding of RTL design, HDL simulation and timing analysis for digital circuits.

Web Page Development for E-commerce Website — HTML, CSS, JavaScript

- Applied CSS and HTML concepts to design and develop a front-end web page.
- Utilized JavaScript to create a dynamic and interactive web form for enhanced user experience.

TECHNICAL SKILLS

Programming: C++, Verilog HDL, HTML, CSS, JavaScript, Python Programming, MATLAB

EDA Tools: Ansys HFSS, Xilinx Vivado, LTSpice, ModelSim

Simulation and CAD: Solid Edge, HFSS, Proteus, Multisim

Concepts: Digital Logic Design, Antenna Theory, CMOS Basics, BJT/FET Operation, Circuit Simulation

Soft Skills: Public Speaking, Team Leadership, Technical Presentation, Marketing Strategy

CERTIFICATIONS

- VLSI Design Methodology – Maven Silicon
- Microsoft Copilot for Productivity – Microsoft LinkedIn

EXTRA-CURRICULAR ACTIVITIES

- **Event Volunteer – College Fest 2023:** Assisted in managing main stage events and stalls; ensured smooth coordination during peak hours.
- **Design Volunteer – BMSCE Incubation Cell (BIG Foundation):** Contributed to design and branding tasks for startups and incubation events.